

Final Project - Grading Rubric

- **Deliverables:** Project poster, 2-page abstract, a GitHub repository link that includes your complete code, and final presentation.

Evidence and Analysis (30)		Implementation (30)		Delivery of Results (40)	
Relevance to Problem Statement (10)	Depth of Analysis: Reasoning (20)	Use of Appropriate Tools (10)	Correctness (20)	Presentation (10)	Poster + 2-page Abstract (30)

Note:

- *Use of JetStream or other computational resources is not mandatory but is strongly encouraged if needed.*
- *Bonus points will be awarded for implementing Spark (if needed), as it has been introduced in this course.*
- *The key to excelling in the project is showing us a deep understanding of every step you take. If a model fails, provide a clear justification for why it failed. If a model performs well, explain why it works. Your reasoning should be evident - we should be able to follow the thought process behind each decision you make through this project.*

Evidence and Analysis (30 points)

How well do you explore and interpret the data in the context of a real-world problem?

Relevance to Problem Statement (10 pts)

- **Problem Definition:** *Is the real-world problem clearly and precisely stated?*
- **Dataset Choice & Fit:** *Does the chosen dataset directly address the problem? Is provenance (source, date, format) documented?*
- **Scope Alignment:** *Are objectives, hypotheses, and evaluation criteria aligned with the problem?*

Depth of Analysis: Reasoning (20 pts)

- **Exploratory Data Analysis:** Quality of summary statistics, visualizations, detection of outliers, missing data treatment.
- **Hypothesis & Logic:** Clear hypotheses, rationale for each analytic step, justification of feature selection/engineering.
- **Statistical Rigor:** Appropriate use of statistical tests, confidence intervals, significance levels where applicable.
- **Insight & Interpretation:** Do findings go beyond “what” to explain “why”? Are patterns and anomalies interpreted in context?
- **Failure Analysis:** If a method/model underperforms, is the cause investigated and alternative approaches considered?

Implementation (30 points)

How technically sound and effective is your approach to solving the problem?

Use of Appropriate Tools (10 pts)

- **Tool Selection:** Justification for languages/libraries (e.g., Python, Spark, Jupyter).
- **Distributed/HPC Resources:** Effective (optional) use of JetStream, cloud VMs, or other HPC.
- **Version Control:** Meaningful Git commits, branch usage, issue tracking or pull requests.
- **Environment Management:** Use of virtual environments, Docker, or similar to encapsulate dependencies.

Correctness (20 pts)

- **Code Quality:** Readability, modularity, adherence to style guides (PEP8, etc.), in-code comments.
- **Functionality:** Code runs end-to-end without errors; handles edge cases and invalid inputs gracefully.

- **Testing & Validation:** Presence of unit tests or sanity checks; reproducible results on provided instructions.
- **Data Handling:** Correct preprocessing (cleaning, normalization), transformation, and postprocessing steps.

Delivery of Results (40 points)

How well key takeaways and findings are presented?

Presentation (10 pts)

- **Oral Delivery:** Clarity of speech, pacing, engagement, ability to answer questions.
- **Slide Design:** Logical flow, readability, appropriate use of graphics/charts, minimal text per slide.
- **Time Management:** Stays within allotted time, allocates time proportional to project components.

Poster + 2-page Abstract (30 pts)

- **Clarity & Conciseness:** Abstract clearly summarizes problem, methods, key results, and conclusions in ≤ 2 pages.
- **Visual Appeal:** Poster layout is well organized, uses headings/legends, readable fonts, effective graphics.
- **Content Coverage:** Poster includes motivation, data description, methodology, main findings, and future work.
- **References & Reproducibility:** Citations for datasets/tools; instructions or links for reproducing results.

Final Course Rubric:

→ Assignments: 40%

→ Assignment Surveys: 10%

→ Final Project: 50%